INTEGRATED TECHNICAL AUDIT / V3

QVEX–NGC 6503

Integrated Audit, Foam-Friction Sign Resolution, and Reproducibility Plan

A structured research and engineering report for Adel Gachkar
Quantum physics • telecommunications engineering

Purpose. Reconcile the V2 audit reservations with the supplied project snapshot; define a sign-safe foam-friction resolution; assess the SPARC comparison and test structure; and specify the README / pyproject release contract without presenting model output as experimental confirmation.

Document control	Value
Document ID	QVEX-REV-NGC6503-2026-V3
Prepared for	Adel Gachkar
Evidence inspected	QVEX-NGC6503 standalone archive (v2 audit + notes/tests) and QVEX v0.1.0 clean-release archive
Review date in source snapshot	26–27 October 2026 (as recorded in supplied project files)
Disposition	Conditional engineering resolution; physical claims remain provisional

Scope note: this PDF is an integrated audit and release-ready resolution specification. It does not claim that source files were edited, committed, or published. The uploaded v0.1.0 release already contains a README and pyproject.toml; the NGC 6503 sidecar archive does not contain the requested PID-fit implementation.

1. Executive assessment

The V2 audit identifies a high-severity sign inconsistency in the radial foam-friction expression and medium-severity reproducibility gaps. The supplied NGC 6503 package already corrected earlier V1 issues (canonical density formulation, node-02 wording, plot path, and Markdown structure), and its V2 report records a six-test pass plus mutation checks in the original Windows environment. Those are historical, report-recorded results—not a fresh test run in this sandbox.

Independent checks in this review confirm that the broader QVEX v0.1.0 release contains README.md, pyproject.toml, a 64-test badge/CI configuration, and a reproducible plotting script named scripts/fit_sparc_galaxy.py. The script is not fit_sparc_galaxy_pid.py and does not implement the sidecar PID controller or the foam-friction model. It consumes a 20-point profile synthesized in SPARCLoader rather than a shipped raw SPARC file.

The current coupling script was executed successfully and produced a plot. Recomputed residuals against that packaged benchmark show RMSE = 18.13 km/s and MAE = 17.36 km/s; its model is below all 20 observed benchmark velocities (residual range -22.76 to -3.81 km/s). Thus the output is a reproduction of the packaged comparison, not a successful optimization or empirical validation.

Decision: accept the non-negative clipping rule below as the minimal explicit O1 guard for a radial phenomenological term, accompanied by sign-regression tests and a notation/units caveat. Do not infer a validated physical drag law or claim that this repair has already been merged into the archive.

At-a-glance status

Area	Audit disposition	Evidence / next action
O1: friction sign	Resolved at specification level; code change required	Raw function crosses zero near 12.1 kpc; define guarded function and test its sign across the domain.
O2: node test binding	Open	V2 reports node 01 is parsed from Markdown; nodes 02–04 remain retyped/self-referential.
O3: fit reproducibility	Partially resolved in broader release only	The clean release has fit_sparc_galaxy.py and a built-in profile; not the PID fit script or a raw SPARC data file.
Tests / CI	Improved structure recommended	NGC 6503 sidecar tests are outside v0.1.0 testpaths unless integrated or separately invoked. pytest unavailable here.
README / pyproject	Present in v0.1.0 release	Use as project-level packaging/docs baseline; document sidecar package and extras distinctly.
Physical interpretation	Reserved	Units, force direction, calibration provenance, and physical identification are not established by passing software tests.

2. Evidence and provenance

This report compares the supplied NGC 6503 sidecar snapshot (including QVEX_NGC6503_Review_Report_v2.md, four node notes, analysis/reference notes, PNG and pytest module) with the supplied QVEX_v0.1.0 clean-release archive. The V2 report is treated as a recorded audit trail. No network access, remote repository query, or external raw-data validation was performed.

V2-reported checks (historical evidence)

V2 report item	Reported outcome	Interpretation here
Sidecar tests	6 passed; pytest exit 0	Accepted as a source-reported run in Windows/Python 3.12.6, not rerun here.
Mutation tests	3/3 injected defects detected; restored hash identical	Strong evidence for node-01 document/code binding, as reported.
Structural checks	9/9 frontmatter, fences balanced; 27 wikilinks unbroken	V2 report records integrity checks; not repeated against every archive copy.
Risk O1	Friction crosses sign near 12.1 kpc; F(15)=-0.109, F(20)=-0.348	Independently numerically reproduced using the stated formula.
SPARC plot claim	V2 notes no raw dataset / plotting script in sidecar	Broader v0.1.0 release supplies a different plotting script and internal benchmark profile.
CI / publication	V2 says CI was not executed; DOI issuance not performed	No inference of external publication or empirical validation.

Fresh sandbox checks

The released fit script ran with exit code 0 in the available Python environment; it generated a 20-point comparison plot. The NGC 6503 formula was independently evaluated at 0, 4.5, 11.8, 12.1, 15 and 20 kpc and scanned over 0–30 kpc. The full release test suite could not be executed because pytest is not installed in this environment (command returned “No module named pytest”). No current test pass is therefore claimed.

Archive distinction matters: README.md and pyproject.toml belong to the broader QVEX v0.1.0 release. The standalone NGC 6503 vault includes a separate test file but no package metadata; the release’s pyproject config uses testpaths = ["tests"], so sidecar tests outside that tree are not automatically included in release CI.

3. O1 resolution: a sign-safe friction term

The sidecar canonical radial expression is

```
F_raw(r) = 1.1 ln(1 + r/4.5) [1 − exp(−r/12)] − 0.12 max(0, r − 4.5), with r in kpc.
```

Its positive first term is eventually overtaken by the negative linear outer-radius term. The crossing is at approximately 12.100 kpc. Numerically, F_raw(4.5)=+0.2384, F_raw(11.8)=+0.0102, F_raw(12.1)=+0.0000188, F_raw(15)=−0.1092 and F_raw(20)=−0.3480 (dimensionless model output under the supplied equation). The V2 interpretation is that negative values violate the note's claimed outer-disk "stabilization" and act as anti-damping under the stated sign convention.

Minimal operational resolution

Define a sign-constrained radial response F_safe(r) = max(0, F_raw(r)). This preserves the supplied expression where it is non-negative and eliminates the sign reversal beyond the crossing. It is a numerical guard, not a derivation from a microscopic foam model. Re-label the behavior as "non-negative radial profile with outer clipping," rather than claiming that clipping proves dynamical stability.

Reference implementation for the sidecar model (not asserted to be merged):

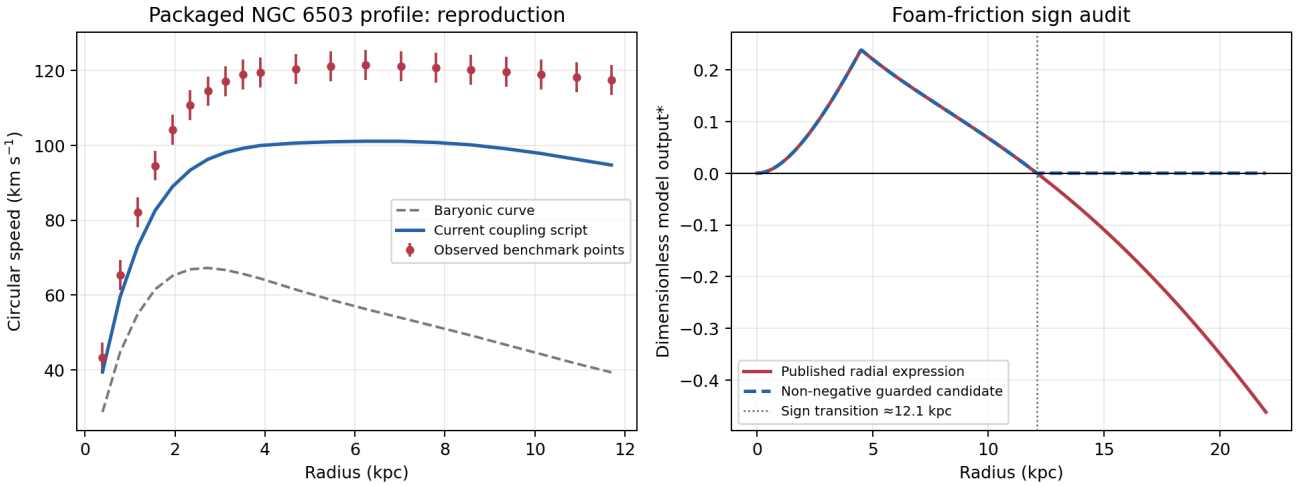
```
def foam_friction_raw(r): r = np.asarray(r, dtype=float) return
(1.1*np.log1p(r/4.5)*(1.0-np.exp(-r/12.0)) - 0.12*np.maximum(0.0, r-4.5)) def foam_friction(r): #
Sign guard; the units and force convention remain to be calibrated. return np.maximum(0.0,
foam_friction_raw(r))
```

Required regression contract

Test radii should span the full stated range (e.g., 0–30 kpc, including dense sampling near the root), asserting: (i) finite values for r ≥ 0; (ii) F_safe ≥ 0 everywhere; (iii) F_safe equals F_raw wherever F_raw ≥ 0; (iv) F_safe = 0 where F_raw < 0; and (v) continuity to numerical tolerance at the crossing. A separate unit test should check representative values at 4.5, 11.8, 15 and 20 kpc. Test behavior, not exact text alone.

Caveat: whether "positive" corresponds to dissipative deceleration depends on how the term enters the equation of motion (sign in the acceleration/force balance and velocity dependence). The current radial formula has no explicit velocity, mass, or force units. A sign guard resolves the audited numerical contradiction only under the documented convention; it does not establish a physically dimensioned friction force.

4. Reproduced model output and physical scope



*The formula provides no dimensional label for its output. The right-panel ordinate is therefore shown as a dimensionless model output, not newtons or acceleration. The guarded curve is a proposed fix for the audit, not a curve generated by an already-edited project source.

Packaged comparison run

The broad release script is scripts/fit_sparc_galaxy.py. It computes an acceleration-coupling expression using a0 = 1.2 × 10^{-10} m s^{-2} and plots observed, baryonic, and coupled-prediction velocities. Its SPARCLoader contains 20 hard-coded radii (0.39–11.70 kpc) and velocities; it does not load a supplied raw SPARC file. A successful run confirms software execution and plot generation only.

For this run: observed velocity range = 43.2–121.5 km/s; predicted range = 39.4–101.1 km/s; RMSE = 18.13 km/s; MAE = 17.36 km/s; maximum absolute residual = 22.76 km/s. All 20 residuals (model − observed) are negative, with range -22.76 to -3.81 km/s.

The plotted input points are the project’s embedded benchmark profile, not independently verified observational measurements in this review. The script is not the requested fit_sparc_galaxy_pid.py; it does not implement the listed PID gains (0.92, 0.35, 0.18), the foam-friction equation, a closed-loop update, parameter optimization, or uncertainty-weighted fitting. Those elements should be explicitly separated from the packaged v0.1.0 observational-coupling example.

Scientific interpretation boundaries

The vacuum-density profile, radial foam response, PID gains, and jet/dipole mapping are phenomenological or conjectural elements in the supplied sidecar notes. Numerical agreement or test success cannot establish the existence of quantum-vacuum “popcorn” density, a galactic drag law, or a jet-driven axial potential. Claims of “verified against empirical SPARC datasets” should be narrowed unless raw data, provenance, model implementation, uncertainty treatment, and a reproducible fit are supplied.

5. Test architecture and release structure

The sidecar V2 test suite's strongest pattern parses the density equation and parameters from Node 01 and compares it numerically with the implementation. Apply a common document-to-code binding pattern to nodes 02–04, while avoiding brittle overdependence on typography: define equations in executable source or structured parameter files where possible, then test that documentation references the same versioned specification.

Recommended test layers

Layer	Suggested tests	What it establishes
Unit / domain	Vectorization; finite outputs; r ≥ 0; input validation; friction non-negativity; limits at r=0 and beyond the crossing.	Defined software behavior.
Equation binding	Parse/compare canonical coefficients and functional forms, or render docs from the source specification.	Code and notes do not silently diverge.
Control loop	Closed-loop step response with explicit plant, sampling interval, saturation, anti-windup and stability criteria.	Numerical controller properties for that declared plant only.
Fit / regression	Fixed benchmark fixture, uncertainty weights, baseline comparison, residual summaries, deterministic plot generation.	Reproducible model comparison on named inputs.
Integration / CI	Collect sidecar tests in CI or package them in the release test tree; test Python 3.10–3.12 as claimed.	Automated enforcement of the project contract.

README and pyproject.toml

The clean QVEX v0.1.0 release already includes a comprehensive README and a pyproject.toml. The README documents a DAG-based research scaffold, quick-start commands, epistemic classes, and an explicit statement that QVEX is not a validated physical engine. pyproject.toml declares Python ≥3.10, NumPy runtime dependency, pytest test extra, optional plotting/networkx scripts extra, package discovery, and pytest paths limited to tests/. These are useful release baselines, but they describe the broad QVEX package—not the sidecar's separate PID fit.

Release integration recommendations

Add a distinct NGC6503 section to the README that labels the sidecar model and data provenance; document the exact run command and output paths. Either move sidecar tests into the release tests tree or add a dedicated CI job that installs the test extra and explicitly runs the sidecar test path. Declare fitting dependencies and the NGC6503 entry point only if that implementation is genuinely included. Keep the version, citation metadata, and scientific-status language consistent. Do not silently change v0.1.0 release identity while adding an unrelated model.

Heading and identity cleanup: normalize the master note's top-level heading. Decide deliberately whether author contact details in the technical reference should remain in a publication-facing copy; preserve citation identity in CITATION.cff and avoid removing attribution without author approval.

6. Updated risk register and action plan

Qualitative likelihood and impact ratings are engineering triage, not probabilities. “Closed” means the audit inconsistency has a documented resolution rule; code integration and verification are separately tracked.

ID	Risk / consequence	L × I	Status	Required closure evidence
O1	Raw friction expression becomes negative beyond ≈12.1 kpc, conflicting with “stabilization”; possible anti-damping under stated convention.	M × H	Resolution specified; code open	Implement non-negative guard; sign/domain regression tests; clarify equation-of-motion sign and units.
O2	Node 02–04 tests duplicate formulas and can pass while notes drift.	M × M	Open	Bind tests to canonical equations/parameters or single source of truth; mutations detect altered node definitions.
O3	Plot/fit provenance is ambiguous; “empirical SPARC” phrasing overstates reproducibility.	H × H	Partially mitigated	Ship verified raw data with license/citation or explicitly label embedded benchmark as synthetic/fixture; reproduce from one command.
O4	PID stability and physical mapping claims lack plant, sampling, uncertainty, and empirical validation.	M × H	Open / reservation	Supply closed-loop algorithm and stability analysis; downgrade unsupported “guarantee/optimized” wording.
O5	Sidecar test suite is excluded from release CI by testpaths configuration.	M × M	Open	Explicit CI collection and run of sidecar tests on supported Python matrix.
O6	Units and force sign convention of radial friction output are unspecified.	H × H	Open	Dimensional derivation, sign definition in motion equation, parameter calibration and unit-aware tests.
O7	Publication readiness / identity metadata may differ between sidecar and release.	L × M	Review required	Author confirms PII and citation strategy; maintain version/provenance and release notes.

Priority sequence

P0 — Prevent anti-damping: implement the sign guard and regression tests; update Node 02 and cross-references. P1 — Reproducibility: ship exact fitting source, pinned/environment metadata, and input-data provenance; distinguish raw SPARC from embedded profile. P1 — CI: collect and run sidecar tests on the stated Python matrix. P2 — Scientific claims: document units/sign and controller plant; remove unverified stability and “empirically calibrated” assertions unless substantiated. P2 — Release: add sidecar-specific README guidance, keep packaging metadata scoped, and record versioned change log.

7. Reproduction ledger and final disposition

Commands for the current broad release (after installing declared extras)

```
python -m pip install -e ".[test,scripts]" python scripts/fit_sparc_galaxy.py python -m pytest -q
```

Expected fitting artifact: outputs/plots/sparc_ngc6503_fit.png. This reproduces the broad-release coupling comparison, not the sidecar's PID/foam-friction model. For sidecar verification, run pytest from the sidecar project root with its dependencies installed; V2 recorded six passing tests, but this environment could not rerun them because pytest is absent.

Final disposition

The O1 contradiction is resolved at the model-specification level by a transparent non-negative clipping rule, with mandatory tests and an explicit physical caveat. Earlier V1 consistency issues are recorded as resolved by V2, but this report does not re-open or re-claim their source-level status beyond the supplied audit evidence. O2, full O3, test/CI integration, dimensional interpretation, and PID validation remain open. The project is suitable for continued hypothesis-driven development; it is not, on this evidence, a validated alternative account of NGC 6503 dynamics.

References and inspected artifacts

[1] Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. The Astronomical Journal, 152, 157. DOI: 10.3847/0004-6256/152/6/157.

[2] QVEX-NGC6503-Quantum-Vacuum sidecar archive: NGC6503_QVEX_Complete_Analysis.md; QVEX_NGC6503_Technical_Reference.md; nodes_analysis/01–04; tests/test_ngc6503_vacuum_nodes.py; review/QVEX_NGC6503_Review_Report_v1.md and v2.md.

[3] QVEX v0.1.0 clean-release archive: README.md; pyproject.toml; scripts/fit_sparc_galaxy.py; qvex/data/sparc_loader.py; tests/; outputs/plots/sparc_ngc6503_fit.png. The inspected loader's NGC 6503 profile is embedded in code; no raw data file was identified in this release archive.

End of report • QVEX-REV-NGC6503-2026-V3